CPT205 Assessment 2: 3D Modelling Project Report

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I. Introduction

With the rapid advancement of computer graphics technology, creating realistic 3D scenes through computer programming has become increasingly feasible. In this project, innovative approaches have been adopted to create an interactive 3D scene. In addition this 3D project implements an interactive driving simulation scene that demonstrates various computer graphics techniques and OpenGL functionality. The scene represents a realistic driving environment where users can control a detailed 3D car model while exploring a carefully designed landscape.

Briefly, the project effectively incorporates a range of graphics techniques to create a comprehensive and interactive 3D scene and the implementation details are as follows:

- **Creation of Geometry:** The scene includes a diverse range of geometric shapes constructed using OpenGL primitives and transformations. Complex structures like playground equipment utilize combinations of basic shapes (cubes, spheres) enhanced through scaling and rotation. Environmental elements such as the lake and garden use polygon-based designs for organic shapes. Detailed objects like the soccer field goals combine line primitives with quad strips to create realistic mesh structures. This approach demonstrates the effective use of OpenGL's geometric primitives to create both architectural and natural elements while maintaining visual quality and performance.
- **Hierarchical Modelling:** The scene is extensively designed using hierarchical modeling techniques, which is evident in several key components. The playground equipment (merry-go-round, seesaw), car model, soccer field goals, and garden elements all demonstrate parent-child relationships in their structure. Specifically, the car model is designed using hierarchical modeling, allowing for complex structures like the car body and wheels to be managed and animated as a single entity. This hierarchical approach significantly enhances code maintainability, animation control, and complex structure manipulation throughout the scene.
- **Transformations:** Various transformations are applied to objects within the scene, including translations, rotations, and scalings, to simulate realistic movements and interactions, such as car steering and wheel rotation.
- **Viewing and Projection:** The project utilizes perspective projection to provide a realistic 3D view and orthogonal projection was also used to draw the interior of the car from a first-person perspective. Also, the scene features multiple camera perspectives, allowing users to switch between free camera movement, third-person car view, and first-person car view.
- **Lighting and Materials:** The scene implements two types of lighting: directional lighting that simulates environmental illumination for both day and night modes, and spot lighting that creates realistic car headlights, with an advanced lighting system is implemented, featuring ambient, diffuse, and specular lighting components to enhance realism. Materials are defined using both default properties for basic objects and custom properties for specific objects to achieve appropriate visual effects.
- **Texture Mapping:** Texture mapping is extensively implemented throughout the scene, with multiple textures applied to enhance realism: ground textures (brick, grass variants), architectural textures (walls, roofs), environmental textures (sky for day/night, lake, highway), and natural textures (leaves, tree trunks). All textures are properly loaded with mipmapping support and appropriate wrapping modes (repeat for continuous surfaces like the ground, clamp for distinct images on buildings).
- **Animation and Interactions:** The scene utilizes dynamic animations and interactive elements: animations (car movement, wheel rotation, steering movements, playground equipment, camera movement, first-person view HUD) and user interactions (car controls, camera switching between multiple views and free camera movement, day/night mode toggling).

Below is a bird's-eye view of the project.

Fig. 1. Overview of the project

II. Design and Features

A. Dynamic Camera System

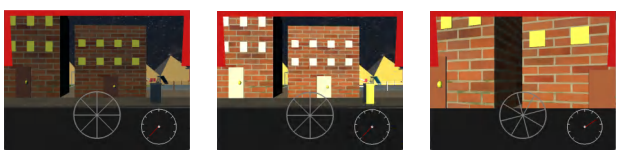
The program implements a sophisticated camera system with multiple viewing modes. Upon program initialization, it defaults to a free camera mode, allowing users immediate freedom to explore the environment:

- Multiple view modes including:

(a) Free camera (b) Third-person view (c) First-person view

Fig. 2. Different viewing modes

  - Free camera mode (default) for exploring the environment
  - Third-person follow mode with dynamic camera positioning
    - Camera automatically maintains optimal distance behind the vehicle
    - Smoothly follows vehicle movement and rotation
    - Provides clear view of the vehicle and surrounding environment
  - First-person driver view with realistic dashboard elements and operational headlights



(a) Light off (b) Light on (c) Animated gauges and steering wheel

Fig. 3. Features under first-person view

    - Functional headlight system for driving
    - Animated gauges and steering wheel
    - Realistic driving perspective
- Smooth transitions between camera modes using interpolation
  - Implements gradual camera movement between mode switches
  - Prevents jarring visual transitions for better user experience
- Interactive camera controls for rotation, zoom, and height adjustment
  - Comprehensive camera manipulation options
  - Adjustable viewing distance and height
  - Dynamic rotation capabilities

(a) Day (b) Night

Fig. 4. Different themes

B. Day/Night Cycle System

A dynamic lighting system that enhances environmental immersion:

- Toggleable day/night modes affecting:
  - Ambient lighting and atmosphere
  - Sky textures and environmental appearance
  - Vehicle headlights functionality
- Dynamic lighting adjustments for enhanced realism
- Headlight system active in first-person view during night mode

C. Environmental Design

The project incorporates diverse modeling elements to create a comprehensive virtual environment. The following sections detail the four main components: Vehicle Components, Road System, Left-side Environment, Right-side Environment and sky elements.

1. Vehicle Components

(a) car head (b) car body (c) Animated wheels

Fig. 5. Some of the car's characteristics

- Detailed red car body with sports car styling and dual headlights mounted on front corners
- Realistic wheel system with differential steering angles
- Dynamic headlights with interactive illumination on environmental surface
- Realistic first-person view with dashboard, steering wheel, and speedometer
- Advanced physics implementation:
  - Movement distance precisely matches wheel rotation (wheel rotation calculated based on WHEEL_CIRCUMFERENCE)
  - Realistic steering mechanics:
    - Wheels can turn while stationary without affecting car orientation
    - Turning effect only occurs during vehicle movement

    - Opposite steering behavior when reversing
    - Wheels gradually return to the center position when steering input stops
  - Natural movement physics:
    - Smooth acceleration when pressing forward
    - Progressive deceleration due to friction when no input is given
    - Maximum speed limit to maintain realistic behavior
    - Reduced turning radius at higher speeds for improved stability

2. Road System

(a) Driveways and sidewalks (b) Realistic visuals (c) Lamp post, benches and trash bins

Fig. 6. Some of the car's characteristics

- Textured highway with realistic surface
- Lamp posts along both sides
- Strategically placed trash bins for environmental detail
- Benches positioned for rest areas
- Pedestrian walkways with distinct texturing
- Perspective projection (gluPerspective) implementation creates realistic depth perception:
  - Objects appear larger when closer to the viewpoint as shown in above graphs
  - Objects gradually diminish in size with distance
  - Creates natural depth cues for improved spatial awareness

3. Left-side Environment

III. Environmental Features

- Natural lake with transparent water effects

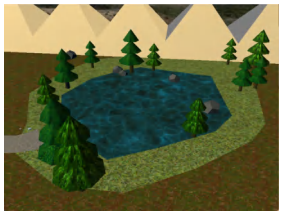
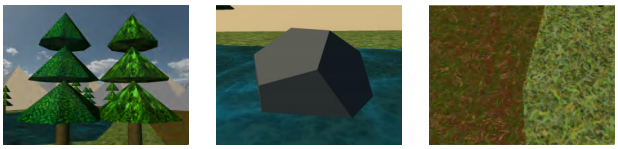


Fig. 7. Transparent water effects with reflection

- Various types of trees with detailed trunk and leaf textures
  - Tree structure generated through hierarchical composition:
    - Cylindrical trunk with realistic bark texture mapping
    - Multiple texture variants for visual diversity
  - Height and trunk radius parameters allow for natural size variations
- Decorative rocks around the lake
- Multi-textured grass areas



(a) Trees with different texture (b) Lake-side decorative rocks (c) Multi-textured grass implementation

Fig. 8. Environmental details showing trees, rocks, and grass texturing

- Static fighter jet display
- Winding garden paths with decorative small stones (implemented using Bézier curves for smooth path generation)
- Flower garden featuring multiple flower types: daisy, rose and tulip.

(a) Static fighter jet model (b) Paths with stone details, generated using Bézier curve algorithms (c) Flower garden with variety of flower types

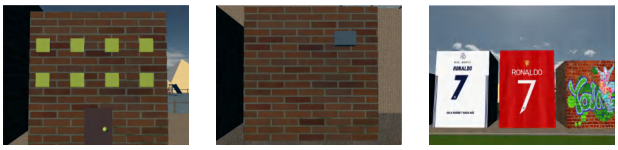
Fig. 9. Other elements

4. Right-side Environment

The right side of the environment contains several key elements that create an urban recreational area:

1) Multi-story Buildings: The buildings feature realistic architectural elements:

- Textured surfaces using clamped texturing to prevent distortion
- Multiple windows arranged in a grid pattern
- Realistic doors with frames and handles
- Rooftop air conditioning units



(a) Front of buildings (b) Roof (c) Wall graffiti

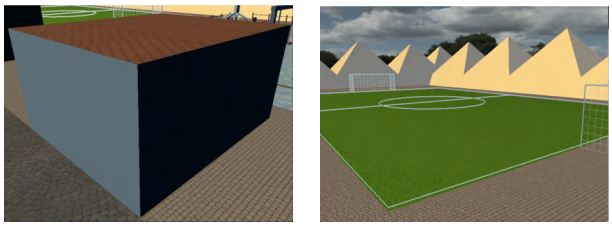
Fig. 10. Building elements

2) Garage Structure: A simple standalone garage structure.

3) Soccer Field: A regulation-style soccer field including:

- Full-size goal posts at both ends
- Standard field markings:
  - Center circle

- Penalty areas
- Corner arcs
- Touchlines and goal lines

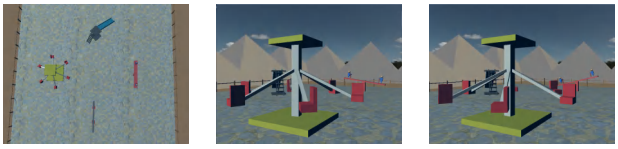


(a) Garage (b) Football court

Fig. 11. Decorative elements

4) Playground Area: A comprehensive children's playground featuring:

- Simple swing sets
- Simple slides
- Seesaw
  - Animation
  - Comfortable seating
  - Handle grips
- Merry-go-round
  - Animation
  - Central rotating platform
  - Multiple seating positions
- Protective Fencing
  - Surrounds entire playground area
  - Appropriate height for child safety
  - Gated entrance points



(a) Overview (b) Dynamic elements (c) Dynamic elements2

Fig. 12. Playground

All elements are implemented using OpenGL primitives and appropriate texturing techniques to create a realistic and engaging environment. Special attention is paid to proper scaling and positioning to maintain a cohesive scene layout.

IV. Instruction

The control instructions will be explained in the following sections: initial free camera perspective control, switching to third-person perspective locked on the car, and switching to first-person perspective. (All the keyboard controls are case-insensitive)

A. Global environment controls

- **M key**: Toggles between day and night modes, accessible from all viewing perspectives. When activated:
  - Day mode features bright ambient lighting and a blue sky texture
  - Night mode reduces ambient lighting intensity and switches to a starry sky texture
  - The transition affects the entire environment's lighting conditions and atmosphere

Users can press the 'M' key to toggle between day/night modes at any time.

B. Free camera controls

When the program starts, it defaults to free camera mode. In this mode:

- **W, S, A, D keys**: Control camera movement
  - **W**: Move camera forward
  - **S**: Move camera backward
  - **A**: Move camera left
  - **D**: Move camera right
- **Arrow keys**: Control camera rotation
  - **Up/Down**: Adjust vertical rotation angle
  - **Left/Right**: Adjust horizontal rotation angle
- **H ,L keys**: Control camera height
  - **H**: Make the camera rise
  - **L**: Make the camera go down

C. Third-Person controls

When in free camera mode, pressing the 'C' key switches to a third-person perspective locked on the vehicle. In this mode:

- **W/S/A/D**: Controls the vehicle's movement
  - W: Accelerate forward
  - S: Brake/Reverse
    - The vehicle has a maximum speed limit
    - When not accelerating, the vehicle will gradually decelerate to a stop due to simulated friction
  - A/D: Turn left/right
- **I/O**: Controls camera zoom
  - I: Zoom in
  - O: Zoom out
- **Up/Down Arrow Keys**: Adjusts the camera's vertical angle

Pressing 'C' again will exit the third-person perspective mode and return to free camera mode.

D. First-Person controls

While in third-person perspective mode, pressing the 'V' key switches to first-person perspective, where users can simulate the actual driving experience. In this mode:

- The steering wheel rotates realistically with turning movements
- The speedometer displays actual vehicle speed
- Vehicle controls:

  - **W**: Accelerate
  - **S**: Brake/Reverse
  - **A/D**: Turn left/right
  - **T**: Toggle headlights on/off

  In this mode, users can return to third-person view by pressing 'V' again, or switch to free camera mode using the 'C' key.
- Perspective controls:
  - **V**: Return to third-person perspective
  - **C**: Switch back to free camera mode

E. Additional Features

- Real-time speedometer display in first-person view
- Dynamic steering wheel animation that responds to turning
- Realistic car window frame visualization in first-person mode
- Smooth camera transitions between different viewing modes

F. Physics Simulation

The program implements basic vehicle physics including:

- Gradual acceleration and deceleration
- The car is affected by ground resistance
- Realistic turning radius at different speeds
- Collision detection with mountain boundaries

These controls allow users to fully explore the 3D environment while experiencing realistic vehicle behavior and multiple viewing perspectives. These interactions can be organized into the following table:

TABLE I
INTERACTIVE CONTROLS

Control	Function
Global Controls	
M key	Day/Night mode switching
Free Camera Mode	
Arrow Up	Rotate camera up
Arrow Down	Rotate camera down
Arrow Left	Rotate camera left
Arrow Right	Rotate camera right
H key	Increase camera height
L key	Decrease camera height
W key	Move camera forward
S key	Move camera backward
A key	Move camera left
D key	Move camera right
Third-Person controls	
C key	Toggle Free camera view/third person view
W key	Accelerate forward
S key	Brake/Reverse
A key	Turn left
S key	Turn right
Arrow Up	Rotate camera up
Arrow Down	Rotate camera down
Arrow Left	Rotate camera left
Arrow Right	Rotate camera right
I key	Zoom in
O key	Zoom out
First-Person controls	
V key	Toggle third/first person view
L key	Switch on/off headlights
W key	Accelerate forward
S key	Brake/Reverse
A key	Turn left
S key	Turn right

V. CONCLUSION

This project successfully implements an interactive 3D driving simulation that effectively demonstrates comprehensive understanding and application of computer graphics techniques. The implementation excels in meeting key assignment objectives:

A. Scene Design and Content(creation of geometry)

- Created a well-structured 3D urban environment representing daily life, featuring roads, buildings, parks, and natural elements
- Implemented both static objects (buildings, mountains, trees) and dynamic elements (controllable vehicle, animated components)
- Achieved realistic scene composition through careful placement and scaling of objects

B. Graphics Techniques Implementation

- **Hierarchical Modeling:** Demonstrated in complex object structures such as the vehicle model with articulated wheels and steering mechanisms
- **Transformations:** Utilized matrix transformations for object positioning, rotation, and animation
- **Viewing and Projection:** Implemented multiple camera modes (free, third-person, first-person) with smooth transitions
- **Lighting and Materials:** Created dynamic lighting system with day/night cycles and realistic material properties
- **Texture Mapping:** Applied diverse textures across multiple surfaces with proper coordinate management
- **Animation and Interaction:** Developed responsive user controls and smooth object animations

C. Technical Excellence

- Effective utilization of freeglut and OpenGL functions for all graphics operations
- Well-structured code with clear organization and comprehensive documentation
- Robust error handling for resource management and runtime operations
- Optimized performance through efficient use of graphics techniques

The project successfully demonstrates mastery of computer graphics principles through practical implementation, creating an engaging and visually impressive interactive environment while maintaining high technical standards and good programming practices.